C-AR.1.1 Check

Simplify linear expressions

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 5

Learning goal: I can add and subtract linear expressions. 5 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$\left(3+\frac{5}{6}p\right)+\left(\frac{5}{6}p-1\right)$$

A. $\frac{5}{6}p+2$

B. $\frac{5}{3}p+4$

C. $\frac{5}{3}p+2$

D. $\frac{11}{3}p$

Circle one letter.

2 Which expression is equivalent to the expression shown?

$$(0.4x+2)+(1.6x-2.5)$$

A. $2x-0.5$

B. $1.5x$

C. $0.64x-0.5$

D. $2x+0.5$

Circle one letter.

3 Which expression makes the equation true?

$$\left(\frac{2}{3}x+6\right)-(__)=-\frac{3}{8}x-2$$

A. $\frac{25}{24}x+8$

B. $\frac{7}{24}x+8$

C. $\frac{25}{24}x+4$

D. $-\frac{25}{24}x-8$

Circle one letter.

4 Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$\left(\frac{4}{3}x - 8\right) - \left(\frac{7}{3}x + 4\right)$$

A. $-x - 4$

B. $x - 12$

C. $\frac{11}{3}x - 4$

D. $-x - 12$

Circle one letter.

5 A rectangle has a length of $\left(\frac{5}{2}x + 0.5\right)$ meters and a width of $(x - 2)$ meters.

Which expression gives the perimeter of the rectangle, in meters, written with the fewest possible terms?

A. $7x - 1.5$

B. $\frac{7}{2}x - 1.5$

C. $7x + 5$

D. $7x - 3$

Circle one letter.

Last question: look at the scale. Which level are you at right now?

LEARNING GOAL I can add and subtract linear expressions.

4	I can write and simplify an expression from a real-world situation.
3	I can add and subtract linear expressions with rational coefficients.
2	I can combine like terms with integer coefficients.
1	I need help to start this.

Circle one: **Level 1** **Level 2** **Level 3** **Level 4**

C-AR.1.1 Re-check

Simplify linear expressions

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 3

3 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$(6x - 6) - (0.49x - 5 + 0.58x)$$

A. $4.93x - 11$

B. $6.09x - 11$

C. $7.07x - 1$

D. $4.93x - 1$

Circle one letter.

2 Which expression is equivalent to the expression shown?

$$(3.2x + 2) + (4.5x - 0.5)$$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

Circle one letter.

3 Which expression makes the equation true?

$$\left(\frac{3}{2}x + 8\right) - (__) = -\frac{5}{3}x - 1$$

A. $-\frac{19}{6}x - 9$

B. $\frac{19}{6}x + 7$

C. $-\frac{1}{6}x + 9$

D. $\frac{19}{6}x + 9$

Circle one letter.

C-AR.1.1 Review

Simplify linear expressions

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 2

2 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$(2x - 4) - (0.59x - 1 + 0.44x)$$

A. $3.03x - 3$

B. $0.97x - 5$

C. $1.85x - 5$

D. $0.97x - 3$

Circle one letter.

2 Which expression makes the equation true?

$$\left(\frac{5}{3}x - 7\right) + (__) = \frac{7}{4}x - 2$$

A. $\frac{1}{12}x - 9$

B. $-\frac{1}{12}x - 5$

C. $\frac{1}{12}x + 5$

D. $\frac{41}{12}x - 9$

Circle one letter.

C-AR.1.2 Check

Determine if two expressions are equivalent

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 5

Learning goal: I can tell whether two expressions are equivalent. 5 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Select **all** the expressions that are equivalent to $\frac{5}{4}(12x + 20)$.

*Select **all** the correct answers.*

A. $\frac{5}{4}(12x) + \frac{5}{4}(20)$

B. $\frac{53}{4}x + \frac{85}{4}$

C. $15x + 25$

D. $15x + 20$

E. $40x$

Circle every correct letter.

2 A student rewrote $\frac{4}{3}(12x + 12)$ as $28x$.

The correct equivalent expression is ____.

A. $\frac{40}{3}x + \frac{40}{3}$

B. $16x + 16$

C. $16x + 12$

D. $9x + 9$

Circle one letter.

3 Are the two expressions equivalent?

$3.5(2x + 4) - 5x + 1$ and $2x + 15$

A. No, because the first simplifies to $12x + 15$, not $2x + 15$.

B. Yes, because both of them simplify to $2x + 15$.

C. No, because $3.5(2x + 4)$ is $7x + 4$, giving $2x + 5$.

D. Yes, because both have an x-term and a constant term.

Circle one letter.

4 Are the two expressions equivalent?

$2.5(8x - 2) + 4x + 8$ and $24x + 3.5$

A. Yes, because both of them simplify to $24x + 3.5$.

B. No, because $2.5(8x - 2)$ is $20x - 2$, giving $24x + 6$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.

Circle one letter.

5 Two expressions are shown.

$4(x + 2) - 3x$ and $2(3x + 4) - 4x$

Are the two expressions equivalent? Choose the answer with the correct reason.

A. Yes; both have an x-term and a constant term of 8.

B. No; the first simplifies to $x + 8$ and the second to $2x + 4$.

C. Yes; both equal 8 when $x = 0$, so they are equivalent.

D. No; the first simplifies to $x + 8$ and the second to $2x + 8$, so the x-terms are different.

Circle one letter.

Last question: look at the scale. Which level are you at right now?

LEARNING GOAL I can tell whether two expressions are equivalent.

4	I can show two expressions are equivalent using the properties of operations and explain which property I used.
3	I can decide whether two linear expressions are equivalent and rewrite one as the other.
2	I can use the distributive property on a simple expression.
1	I need help to start this.

Circle one: **Level 1** **Level 2** **Level 3** **Level 4**

C-AR.1.2 Re-check

Determine if two expressions are equivalent

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 3

3 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Are the two expressions equivalent?

$1.5(4x - 6) + x + 8$ and $7x - 2$

A. No, because $1.5(4x - 6)$ is $6x - 6$, giving $7x + 2$.

B. Yes, because both of them simplify to $7x - 2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x - 1$, not $7x - 2$.

Circle one letter.

2 Which expression is equivalent to $2.5(6x - 4)$?

A. $15x - 4$

B. $15x - 10$

C. $11x$

D. $8.5x - 1.5$

Circle one letter.

3 Select **all** the expressions that are equivalent to $\frac{3}{2}(6x + 2)$.

*Select **all** the correct answers.*

A. $\frac{3}{2}(6x) + \frac{3}{2}(2)$

B. $\frac{15}{2}x + \frac{7}{2}$

C. $12x$

D. $9x + 2$

E. $9x + 3$

Circle every correct letter.

C-AR.1.2 Review

Determine if two expressions are equivalent

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 2

2 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. No calculator.

1 Which expression is equivalent to $\frac{2}{3}(9x - 15)$?

A. $6x + 10$

B. $6x - 15$

C. $-4x$

D. $6x - 10$

Circle one letter.

2 A student rewrote $\frac{7}{4}(12x + 4)$ as $25x$.

The correct equivalent expression is ____.

A. $\frac{48}{7}x + \frac{16}{7}$

B. $21x + 7$

C. $\frac{55}{4}x + \frac{23}{4}$

D. $21x + 4$

Circle one letter.

C-AR.2.2 Check

Two-step equations with rational numbers

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 5

Learning goal: I can write and solve two-step equations. 5 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 A park charges a $25.00 entry fee plus $12.50 for each hour of skate rental. A visitor paid $62.50 in all.

Which choice shows an equation that can be used to find h, the number of hours of skate rental the visitor paid for, and its solution?

A. $12.5h - 25 = 62.5;\ h = 7$

B. $25h + 12.5 = 62.5;\ h = 2$

C. $12.5h + 25 = 62.5;\ h = 7$

D. $12.5h + 25 = 62.5;\ h = 3$

Circle one letter.

2 Solve the equation for h.

$$1.5(h + 1.6) = -0.6$$

Write a number only. No units, $ or %.

Answer: _____

3 Three students each saved the same amount, s dollars, and then each earned $5.00 more. Together they now have $105.00.

Which choice shows an equation that can be used to find s, and its solution?

A. $3(s + 5) = 105;\ s = 30$

B. $3s + 5 = 105;\ s = 33\frac{1}{3}$

C. $3(s + 5) = 105;\ s = 35$

D. $3(s - 5) = 105;\ s = 40$

Circle one letter.

4 What is the exact value of g in the equation shown?

$$-1\frac{7}{9} = \frac{2}{3}g - 4$$

Write a number only. No units, $ or %.

Answer: _____

5 Gym A charges $24.50 each month. Jordan used a coupon worth $35.00 off the total and paid Gym A $185.50 in all.

Let n be the number of months Jordan paid for. Gym B charges $26.50 each month and takes no coupons. Which choice gives n, the gym that would have cost less for the same number of months, and how much less?

A. $n = 6$; Gym B, by $26.50

B. $n = 9$; Gym A, by $53.00

C. $n = 9$; Gym A, by $18.00

D. $n = 8$; Gym A, by $26.50

Circle one letter.

Last question: look at the scale. Which level are you at right now?

LEARNING GOAL I can write and solve two-step equations.

4	I can write a two-step equation from a real-world context, solve it, and check the answer.
3	I can solve a two-step equation, including with rational numbers or the distributive property.
2	I can solve a one-step equation.
1	I need help to start this.

Circle one: **Level 1** **Level 2** **Level 3** **Level 4**

C-AR.2.2 Re-check

Two-step equations with rational numbers

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 3

3 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 Solve the equation for x.

$$-2.5(x - 1.2) = 7$$

Write a number only. No units, $ or %.

Answer: _____

2 Six identical boxes each weigh b pounds with the packaging. After 1.25 pounds of packaging is removed from each box, the 6 boxes weigh 40.5 pounds in all.

Which choice shows an equation that can be used to find b, and its solution?

A. $6(b - 1.25) = 40.5; b = 8$

B. $b - 1.25 = 40.5; b = 41.75$

C. $6(b - 1.25) = 40.5; b = 6.75$

D. $6(b + 1.25) = 40.5; b = 5.5$

Circle one letter.

3 A number, x, is decreased by 8. That result is then multiplied by 3, giving 3.

Which choice shows an equation that represents this situation, and the value of x?

A. $3(x - 8) = 3; x = 9$

B. $3x - 8 = 3; x = 3\frac{2}{3}$

C. $3(x + 8) = 3; x = -7$

D. $3x - 24 = 9; x = 11$

Circle one letter.

C-AR.2.2 Review

Two-step equations with rational numbers

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 2

2 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 A park charges a \$15.00 entry fee plus \$3.75 for each game. A visitor paid \$33.75 in all.

Which choice shows an equation that can be used to find g, the number of games the visitor paid for, and its solution?

A. $3.75g + 15 = 33.75;\ g = 5$

B. $15g + 3.75 = 33.75;\ g = 2$

C. $3.75g + 15 = 33.75;\ g = 13$

D. $3.75g - 15 = 33.75;\ g = 13$

Circle one letter.

2 What is the exact value of t in the equation shown?

$$-5\frac{8}{9} = \frac{4}{3}t - 9$$

Write a number only. No units, \$ or %.

Answer: _____

C-AR.2.1 Check

Solve one-step inequalities

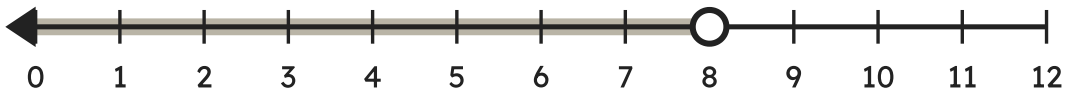
Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 5

Learning goal: I can write and solve one-step inequalities and graph the solutions. 5 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 The graph shows the solution of an inequality.

Which inequality has this solution?



A. $2.5x \leq 20$

B. $x + 2.5 < 20$

C. $2.5x > 20$

D. $2.5x < 20$

Circle one letter.

2 Solve the inequality.

$$-3 + t \geq -6.9$$

What is the solution?

A. $t \geq 3.9$

B. $t \leq -3.9$

C. $t \geq -9.9$

D. $t \geq -3.9$

Circle one letter.

3 Solve the inequality.

$$-0.5n > 0.5$$

Which number line shows the solution?

A. Open circle at –1, shaded left

B. Closed circle at –1, shaded left

C. Open circle at 1, shaded left

D. Open circle at –1, shaded right

Circle one letter.

4 The product of a number x and −0.5 is less than −5.

What are the possible values of x?

A. $x > 10$

B. $x > 2.5$

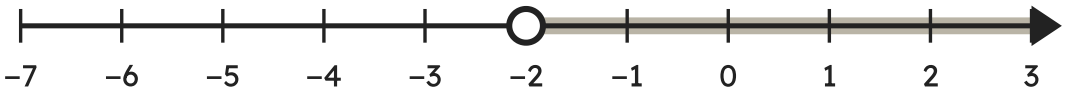
C. $x < -10$

D. $x < 10$

Circle one letter.

5 The number line shows the solution of one of the inequalities below.

Which inequality has this solution?



A. $x + 3.5 \geq 1.5$

B. $\frac{x}{4} < -0.5$

C. $x - 4.5 > -8.5$

D. $-1.5x < 3$

Circle one letter.

Last question: look at the scale. Which level are you at right now?

LEARNING GOAL I can write and solve one-step inequalities and graph the solutions.

4	I can solve a one-step inequality with rational numbers, graph it, and explain why the solution set is correct (no real-world context in this benchmark).
3	I can solve a one-step inequality, including with negative numbers, and graph the solution.
2	I can graph an inequality on a number line.
1	I need help to start this.

Circle one: **Level 1** **Level 2** **Level 3** **Level 4**

C-AR.2.1 Re-check

Solve one-step inequalities

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 3

3 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 The product of a number x and −1.5 is less than −12.

What are the possible values of x?

A. $x > 8$

B. $x > 18$

C. $x < -8$

D. $x < 8$

Circle one letter.

2 Solve the inequality.

$$0.4c \geq -1.2$$

Which describes the graph of the solution on a number line?

A. Closed circle at 3, shaded right

B. Open circle at −3, shaded right

C. Closed circle at −3, shaded right

D. Closed circle at −3, shaded left

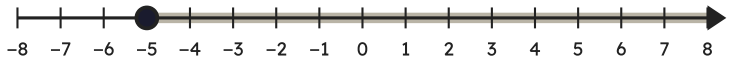
Circle one letter.

3 Solve the inequality.

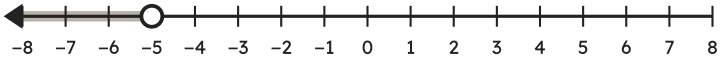
$$\frac{n}{-4} \geq 1.25$$

Which number line shows the solution?

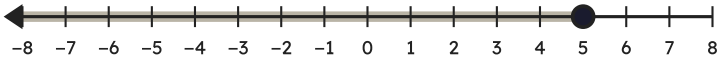
A



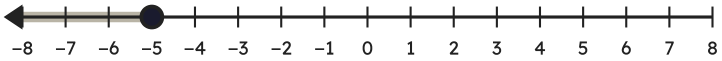
B



C



D



Circle one letter.

C-AR.2.1 Review

Solve one-step inequalities

Name _____ Period _____

Teacher (circle): Mr. C. Shaffer · Mr. J. Shaffer Score _____ / 2

2 questions, 1 point each. Work each one out on paper, then circle the letter or write your answer. Calculator allowed.

1 The product of a number x and −2.5 is at least −20.

What are the possible values of x?

A. $x \leq 50$

B. $x \geq 8$

C. $x \leq 8$

D. $x < 8$

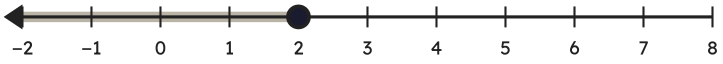
Circle one letter.

2 Solve the inequality.

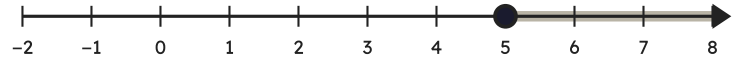
$$0.5w \leq 2.5$$

Which number line shows the solution?

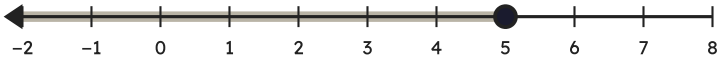
A



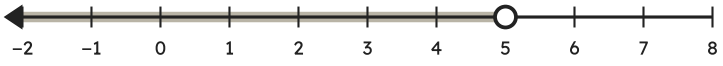
B



C



D



Circle one letter.